



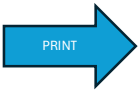
According to the the profitability indicators, here is what can be understood:

- A positive NPV indicates that the project is expected to generate value beyond the cost of capital. In this case, the NPV of 4,740,921 suggests the project is profitable and adds value to the firm.
- A 12% IRR is generally acceptable, but its attractiveness depends on the company's cost of capital. If the cost of capital is lower than 12%, the project is profitable.
- The 9.87-year DPP means that the initial investment will be recovered in present value terms 9.87 years after the start of construction. Since the project starts generating cash flows in Year 2, the DPP implies that the payback occurs 7.87 years after the project becomes operational (i.e., $9.87 - 2 = 7.87$ years). This is a longer payback period than the regular PP, reflecting the impact of discounting and the back-loaded nature of the cash flows.
- Since the project has a 2-year construction phase, the 8.17-year payback period means that the initial investment of 10,000,000 will be recovered 8.17 years after the start of construction. This implies that the project will start generating cash flows in Year 2, and the payback will occur 6.17 years after the project becomes operational (i.e., $8.17 - 2 = 6.17$ years). This is still a relatively long payback period, which may indicate higher risk or lower liquidity, but it is more reasonable when considering the 2-year construction phase.
- A PI greater than 1 indicates that the project is profitable. A PI of 1.47 means that for every dollar invested, the project generates \$1.47 in present value terms. This is a strong indicator of profitability.

The investment is profitable based on the NPV, IRR, and PI. However, the payback and discounted payback periods are still relatively long, even after adjusting for the 2-year construction phase. This suggests that the project may be riskier or less liquid, particularly because the cash flows are back-loaded. If the company's cost of capital is lower than 12% and it is comfortable with the longer payback periods, the project is likely a good investment. Otherwise, the company may need to consider the risk associated with the delayed cash flows.

Break Even Analysis Summary	
Investment	\$ 15,837,697.52
Sales price per bottle	\$ 1.45
Bottles sold after three years	\$ 780,338.11
Inflation Rate	\$ 0.02
Nominal Discount Rate	\$ 0.17

Based on the data in the first image, this break-even analysis evaluates a \$15,837,697.52 investment in what appears to be a bottled product venture. The analysis projects sales of 780,338.11 bottles after three years at a price point of \$1.45 per bottle. With a nominal discount rate of 17% and an inflation rate of 2%, the analysis aims to determine when the cumulative discounted revenue from bottle sales will offset the substantial initial investment. The projected sales volume indicates the estimated quantity needed to reach the break-even point, accounting for the time value of money through the relatively high discount rate.



Break even analysis

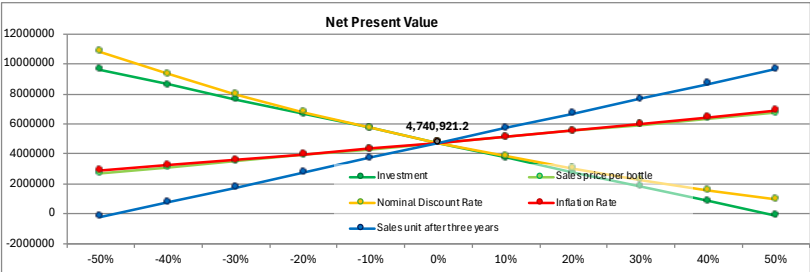
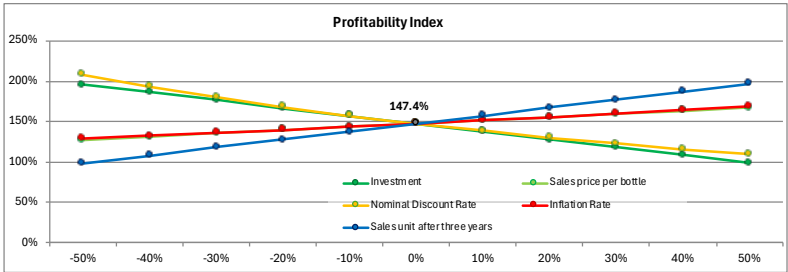
Sensitivity Analysis

This sensitivity analysis illustrates how key variables affect both the Profitability Index and Net Present Value (NPV) of the investment:

The left graph shows the Profitability Index responding to percentage changes in variables. Most notably, increasing the sales units after three years (blue line) significantly improves profitability, rising from about 100% to 200% as values change from -50% to +50%. Conversely, higher sales price per bottle and investment (green lines) show diminishing returns as they increase, while inflation rate (red) gradually increases profitability.

The right graph displays NPV impacts, showing that sales units after three years is the most sensitive variable, swinging from negative to strongly positive NPV (approximately \$9.5 million at +50%). The NPV at baseline is \$4,740,921.2, indicating the project is profitable under current assumptions. Notably, only increasing sales volume consistently improves NPV across all scenarios, while other variables show either diminishing returns or negative effects at higher values.

This analysis suggests the investment's success hinges most critically on achieving or exceeding the projected sales volume after three years, while being relatively less sensitive to changes in other variables.



SCENARIO ANALYSIS

Scenario Summary				
Current Values:		Pessimistic	Realistic	Optimistic
Changing Cells:				
Taxes		20%	25%	20%
production_cost	\$ 0.50	\$ 1.00	\$ 0.50	\$ 0.25
Inflation_rate	4%	8%	4%	-2%
Result Cells:				
NPV	\$4,740,921	\$3,406,981	\$4,740,921	\$1,848,052
IRR	12%	6%	12%	16%
PI	1.47	1.34	1.47	1.18
PP	8.17	10.58	8.17	7.32
DPP	9.87	11.44	9.87	11.19

Notes: Current Values column represents values of changing cells at time Scenario Summary Report was created. Changing cells for each scenario are highlighted in grey.

This scenario summary presents a sensitivity analysis examining how three key variables (taxes, production costs, and inflation rate) affect the investment's financial metrics across pessimistic, realistic, and optimistic scenarios.

The current values (also represented in the realistic scenario) show a favorable NPV of \$4,740,921 with a 12% IRR and profitability index of 1.47, indicating a profitable venture. The pessimistic scenario significantly degrades performance by increasing taxes to 25%, doubling production costs to \$1.00, and raising inflation to 8%. This reduces NPV to \$3,406,981, drops IRR to 6%, and extends the payback period to 10.58 years from 8.17 years.

The optimistic scenario, with lower taxes (15%), halved production costs (\$0.25), and deflation (-2%), improves IRR to 16% but surprisingly reduces NPV to \$1,848,052 and PI to 1.18, suggesting some counterintuitive relationships. The discounted payback period (DPP) increases to 11.19 years in this scenario, which is unexpected for an "optimistic" case.

Overall, while the investment appears generally profitable across all scenarios, production costs and inflation have substantial impacts on performance metrics, and there appears to be an optimal balance rather than a simple linear relationship between more favorable input conditions and better outcomes.

